Hypothesis Testing in a Probabilistic Framework

Intro to probability theory

The magazine Discover once had a special issue on "Life at Risk." In an article, Jeffrey Kluger describes the risks of making it through one day:

> *Imagine my relief when I made it out of bed alive last Monday morning. It was touch and go there for a while, but I managed to scrape through. Getting up was not the only death-defying act I performed that day. There was shaving, for example; that was no walk in the park. Then there was showering, followed by leaving the house and walking to work and spending eight hours at the office. By the time I finished my day – a day that also included eating lunch, exercising, going out to dinner, and going home – I counted myself lucky to have survived in one piece.*

Is this writer unusually fearful? No, why?

Intro to probability theory

- ”There is not a single thing you can do ... that isn’t risky enough to be your last.”
- Examples
  - 1 out of 2 million people will die from falling out of bed.
  - 1 out of 400 will be injured falling out of bed
  - 1 out of 77 adults over 35 will have a heart attack this year
  - Black Americans are twice as likely to be shot by the police compared to Whites
  - The average American faces a 1 in 13 risk of suffering some kind of injury in home that necessitates medical attention
  - 1 out of 32 risk of being the victim of some violent crime, 1 out of 14 risk of having property stolen this year
- These are all probabilities that were calculated from counts of reported accidents
- What is the question of interest from a hypothesis testing framework?

The Fair Coin

- Example: In football, a coin toss$\Rightarrow$ at what point do you become suspicious that the coin is unfair?
- Statistical inference provides a ***systematic way*** of determining risk
- The coin toss can be thought of as a test statistic
  - What is H_0
  - What is H_1
  - What is the Type I error of the test statistic?
  - What is the Type II error?
  - What percentage of the time are you willing to be wrong?

The Fair Coin

- How can we calculate the risk of a Type I error associated with a specific outcome in a test of statistical significance also called the observed significance level?
  - One way is to determine how often a fair coin would be that the Patriots do not always get the ball
    Or my wife and I are not both selected 'at random' for airport 'screening'
    Or there are no black jurors
    We insist on calling these fair simply because we are fooled into believing they are random when they are not…
- Ultimately you are interested in determining the likelihood of getting *X* heads in a very large number of ***samples or trials*** of a fair coin (i.e., tosses in 16 football games for 10 years)
  - The resulting distribution is the sampling distribution
  - Example: The shape of the distribution of the number of events observed in *N* trials is called a binomial distribution

The Logic of Statistical Inference

Hypothesis Testing

Null or **statistical hypothesis**, denoted by H_0

Scientific, alternative, or **research hypothesis**, denoted by H_A or H_1

What is Hypothesis Testing?

Hypothesis testing: A procedure, **based on sample evidence and probability theory**, used to determine whether the hypothesis is a reasonable statement and should not be rejected, or is unreasonable and should be rejected**.** Note: do not ever say "the null hypothesis was accepted in this study…"

Null Hypothesis H_0: A statement about the value of a population parameter that is ***assumed*** to be true for the purpose of testing

- **Example**: Twenty percent of all juvenile offenders are disciplined by the criminal justice system.

Alternative Hypothesis H_a: A statement about the value of a population parameter that is assumed to be true ***if the Null Hypothesis is rejected*** during testing.

- **Example**: The percentage of juvenile offenders who are disciplined is not equal to 20%

Hypothesis Generation

These two hypotheses, designated H_O and H_A (or H_1), are mutually exclusive exhaustive- the don't overlap

Properties of the Null

- Specifies ***no difference*** or no change from a standard or theoretical value
- Always specifies something about a particular ***population*** parameter
- Used in constructing a sampling distribution
- For the subsequent quantitative work, the null hypothesis is assumed to be true

Properties of the Alternative

- Usually stated in general terms
- Mutually exclusive - no overlap with H_O
- Can be directional or non-directional

Directional vs. Non-directional H_As

Non-directional H_A - usually stated as “does not equal” or “is different than”
Directional H_A - stated as “greater than” or “less than”

- Note that a non-directional hypothesis is equal to the two directional hypotheses “greater than” or “less than”

The Null Hypothesis

A statement of no relationship or no difference, denoted H_0

In practice, in statistics, we make decisions about hypotheses in relation to the null hypothesis rather than the research hypothesis

This is because the null hypothesis states that the parameter in which we are interested is a particular value.

- For example, your null hypothesis H_0 might be that there is no difference between sexual minority and heterosexual youth in average number of ACEs, or, put differently, that the difference is equal to zero.

Error in Hypothesis Testing

Whenever we rely on sample statistics to make statements about population parameters, we must always accept that our conclusions are tentative

This is why, when we test hypotheses in research, we generally do not ask whether a hypothesis is true or false. (?)

The question is: can we make an inference, or draw a conclusion, about our hypotheses based on **what we know from a sample**?

Risk of error and Levels of Significance

In statistical inference, we assess the risk of making a wrong decision about the population parameter in reference to Type I error.

- The language is “reject” or “fail to reject” the null
- The amount of error we are willing to risk is called the significance level of a test of statistical significance
- The decision to reject or not is based on a sample statistic
- The significance criterion is denoted alpha α

The estimate of the risk of Type I error that is associated with rejecting the null hypothesis in a test of statistical significance (based on a sample statistic) is called the observed significance level and is ordinarily represented by the symbol *p*.

If $p < \alpha \rightarrow$ reject H_0

Types of Decisions and Errors

	Decision	
State of Nature (reality)	**Fail to reject H_0**	**Reject H_0**
H_0 is true	Correct decision $(1 - \alpha)$	Type I error (α)
H_0 is false	Type II error (β)	Correct decision $(1 - \beta)$ = power

Classic Example: Decision making in the courtroom

Hypo: D is on trial for murder. The jury hears the evidence in the case and must decide guilt or not guilt:

What is H0? Put in terms of “no difference”

What is H1?

Example

- A man is on trial for murder;
- The null hypothesis
- What are the possible decisions

Hypothesis testing framework		
	Person Not guilty	**Person is guilty**
Jury Decides Person Not guilty	Correct	Type I error
Jury Decides Person Guilty	Type II error	Correct

H_0: D is not guilty

H_1: D is guilty

	Decision	
State of Nature	**Fail to reject H_0**	**Reject H_0**
H_0 is true	Correct decision $(1-\alpha)$ D is not guilty (H_0 true) Jury finds D not guilty (DNR H_0) Good result	Type I error (α) D is not guilty Jury finds D guilty Not good result
H_0 is false	Type II error (β) D is guilty Jury finds D not guilty Not good result	Correct decision (1-β) D is guilty Jury finds D guilty Good result

The p-value is the probability of rejecting the null when the null is true, i.e., the probability of a Type I error.

What is the null hypothesis is that some cancer treatment does not work?

Example

	Decision	
State of Nature	**Fail to reject H_0**	**Reject H_0**
H_0 is true	Correct decision $(1-\alpha)$ There is no difference between sexual min. and heterosexual youth, and we conclude that there is no difference	Type I error (α) There is a no difference between sexual min. and heterosexual youth, <u>but we</u> <u>conclude that there is a</u> <u>difference</u>
H_0 is false	Type II error (β) There is a difference between sexual min. and heterosexual youth, and <u>we conclude that</u> <u>there is not a difference</u>	Correct decision $(1\text{-}\beta)$??

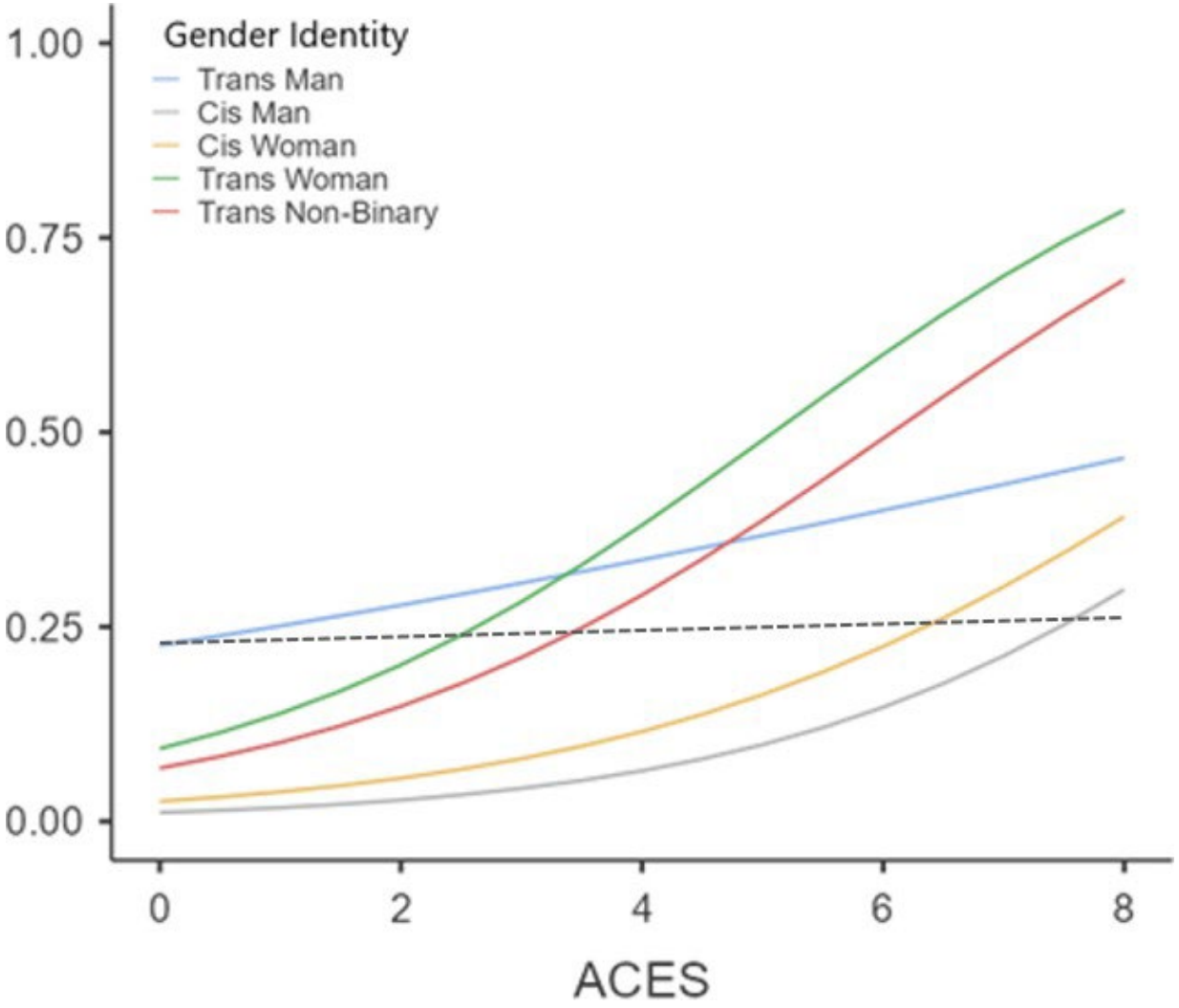

Fig. 2. Predicted probability of suicide attempt by ACEs cumulative risk (CR) across gender identity.

Notes. Predicted probabilities were generated from moderation model of ACEs CR and gender identity controlling for age, race, sexual identity, sex at birth, income, poverty level and education. Dotted line shows similarity between 0 ACEs and 8 ACEs.

What types of conclusions do you draw from this chart in light of what we have been told about ACEs?

Does statistical significance always ‘matter’?

Statistical significance

- Determined through null hypothesis significance testing

Practical significance

- Measured by effect size

Legal significance

- Measured by the weight of the evidence
- Beyond a reasonable doubt
- Probable cause

Type I error

- Building a sampling distribution provides a method for defining our risk of a Type I error
- We cannot construct a sampling distribution every time we want to test a hypothesis, why?
- Fortunately, (or unfortunately) there is another method we can use
- We rely on known sampling distributions based on probability theory (during sampling week)